Name: _____ School: _____

Name: _____ Team Number: _____ Score: _____/92

2019 UT Austin Science Olympiad Regional Tournament

Wifi Lab

Directions: You will have 30 minutes to complete this test. After your exam is complete, you will test your antenna. Complete each question on the test to the best of your ability, and for free response questions, remember to show your work! Good luck!

If you have any questions, comments, concerns, or suggestions about the test, please contact Michael Taleff at mtaleff@utexas.edu.

Section 1. Multiple Choice (1 pt each)

1. What is the approximate speed of light in a vacuum?
   (a) $3.0 \cdot 10^7$ m/s
   (b) $3.0 \cdot 10^8$ m/s
   (c) $3.0 \cdot 10^9$ m/s
   (d) $3.0 \cdot 10^{10}$ m/s
2. What is the wavelength of a photon with a frequency of 533 terahertz.
   (a) $5.62 \cdot 10^{-7}$ m
   (b) $1.60 \cdot 10^{23}$ m
   (c) $1.78 \cdot 10^6$ m
   (d) $7.36 \cdot 10^{-9}$ m
3. What region of the electromagnetic spectrum is a photon with energy $1.18 \cdot 10^{-17}$ joules categorized in?
   (a) Microwave
   (b) Infrared
   (c) Visible
   (d) Ultraviolet
4. As a wave travels through space, a stationary observer records a passing crest every 10 seconds. What is the frequency of this wave in hertz?
   (a) 10 Hz
   (b) 5 Hz
   (c) 0.1 Hz
   (d) 0.2 Hz
5. What is the resistive component of the feed-point impedance of an ideal half-wave dipole antenna?
   (a) 50Ω
   (b) 43Ω
   (c) 73Ω
   (d) 52Ω

6. What is the range of the visible portion of the electromagnetic spectrum?
   (a) 400 to 700 nm
   (b) 400 to 700 μm
   (c) 40 to 70 nm
   (d) 40 to 70 μm
7. What region of the atmosphere reflects radio waves?
   (a) Troposphere
   (b) Mesosphere
   (c) Stratosphere
   (d) Ionosphere
8. What is the energy of a photon with wavelength 22 nm?
   (a) $1.11 \cdot 10^{17}$ J
   (b) $4.37 \cdot 10^{-33}$ J
   (c) $1.00 \cdot 10^{-34}$ J
   (d) $9.03 \cdot 10^{-18}$ J
9. An electromagnetic wave is composed of an oscillating electric and magnetic field. What orientation is this magnetic field to the electric field in an EM wave?
   (a) In-Phase, Perpendicular
   (b) In-Phase, Parallel
   (c) Out-of-Phase, Perpendicular
   (d) Out-of-Phase, Parallel
10. Which of the following is a common radio band used by Wifi?
   (a) 20 THz
   (b) 5.8 THz
   (c) 2.4 GHz
   (d) 15 MHz
11. How does the speed of light in air compare to the speed of light in a vacuum?
   (a) Light travels slower in air
   (b) Light travels faster in air
   (c) Light travels the same speed in air
   (d) Light can travel both faster and slower in air

12. Imagine a photon with a frequency of 300 THz and a photon with a frequency of 300 MHz. If both are traveling in a vaccuum, which is traveling faster?
    (a) Both are traveling at the same speed
    (b) The 300 THz photon
    (c) The 300 MHz photon
    (d) It is impossible to tell
13. A particular wave has a wavelength of 24 m. If a stationary observer observes the crest of that wave passing every 8.0 seconds, what is the velocity of the wave?
    (a) 0.33 m/s
    (b) 0.75 m/s
    (c) 1.5 m/s
    (d) 3.0 m/s
14. Which of the following is used to indicate the speed of information transfer across a radio connection?
    (a) Frequency
    (b) Bauds
    (c) Signal Bandwidth
    (d) Signal-to-Noise Ratio
15. How would the directivity of a parabolic antenna compare to a dipole antenna?
    (a) Both probably have the same directivity
    (b) The dipole has a great directivity
    (c) The dipole has a lower directivity
16. If a half-wave dipole is transmitting in the 2.0 m band, approximately how long is the dipole?
    (a) 8.0 m
    (b) 4.0 m
    (c) 2.0 m
    (d) 1.0 m
17. Choose the correct ordering of frequencies from highest to lowest.
    (a) infrared, microwaves, gamma rays
    (b) gamma rays, microwaves, infrared
    (c) gamma rays, infrared, microwaves
    (d) microwaves, infrared, gamma rays

18. In a Yagi-Uda antenna, which of the following number of directors will result in the highest maximum gain?
    (a) 8 Directors
    (b) 6 Directors
    (c) 4 Directors
    (d) 2 Directors
19. What is the relationship between velocity and frequency in regards to radio waves.
    (a) No Relationship
    (b) Constant Relationship
    (c) Direct Relationship
    (d) Inverse Relationship
20. What is the rearmost element of a Yagi-Uda antenna called (the one furthest from the direction of maximum gain)
    (a) Driven Element
    (b) Reflector
    (c) Director
    (d) Dipole

Section 2. Matching (2 pts each)

_____ Monopole Antenna

_____ Bi-Quad Antenna

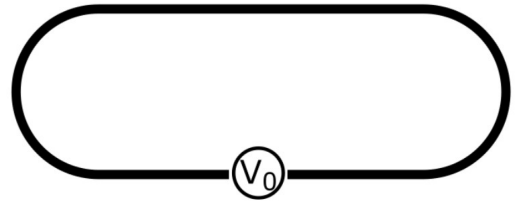
_____ J Antenna

_____ Folded Dipole Antenna

_____ Yagi-Uda Antenna

_____ Dipole Antenna

(u)



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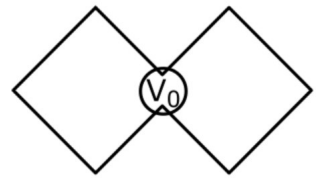

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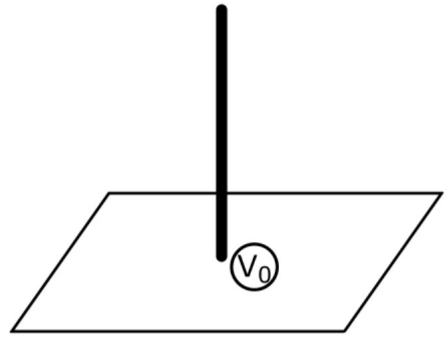

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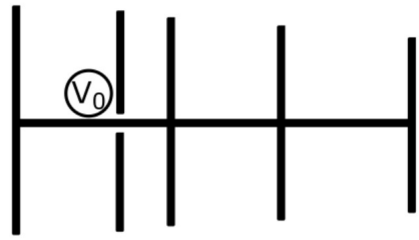


(z)

Section 3. Short Answer (5 pts each)

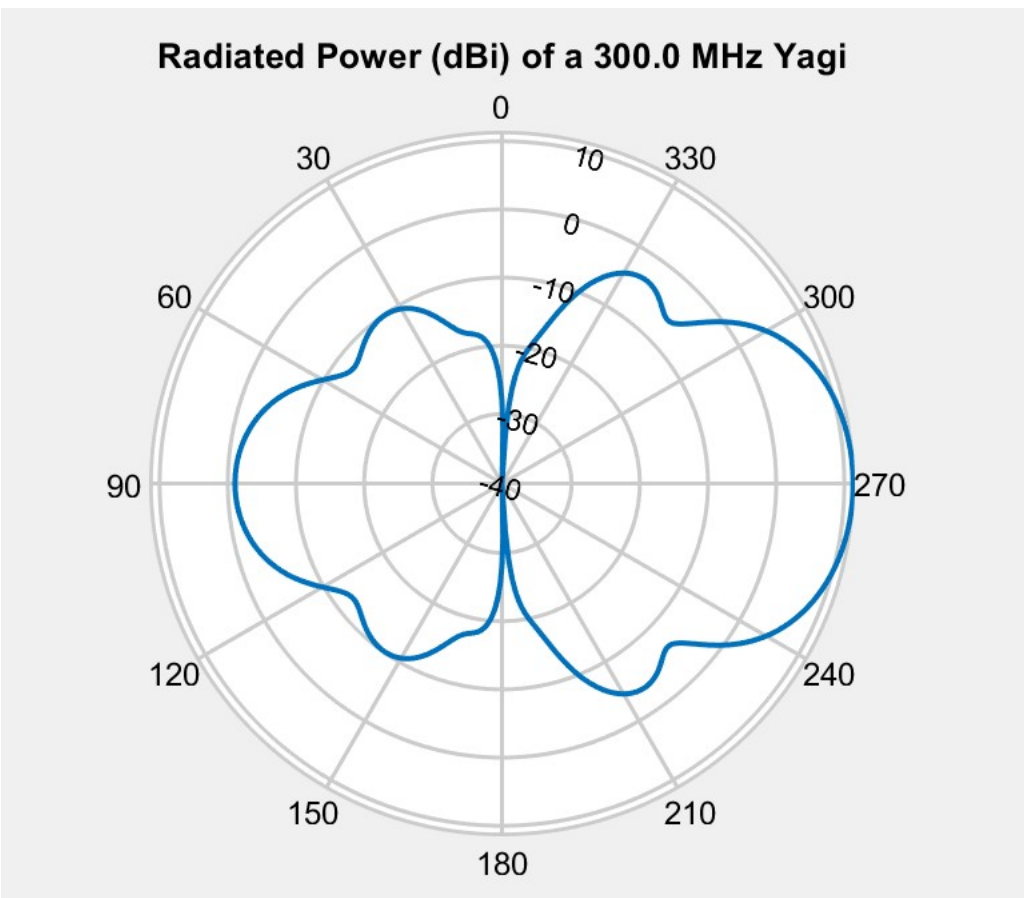
21. List the following antenna types from greatest directivity to least directivity: dipole, isotropic, axial-mode helical.

22. Elmer's half-wave dipole antenna has a directivity of 1.64 and an efficiency of 23.0%. If Elmer wants to radiate 50.0 W of power from his antenna, how much power should he send to his antenna?

23. Define the concept of bandwidth. When using the minimum required bandwidth, would a morse code transmission or an FM voice transmission require more bandwidth and why?

24. When transmitting from his amateur radio station, Elmer always ensures the lowest possible amount of spurious (i.e. unwanted) emissions from his antenna. His rule of thumb is an emitted power of spurious emissions 50.0 dB less than his desired signal. If the emitted power of his desired signal is 100. W, what is the maximum power he can allow for spurious emissions?

25. (*) Below is the far-field radiation pattern for a real Yagi-Uda antenna. Answer the following questions about this radiation pattern. (This question is worth **10 pts**, not 5 pts).



(a) What is the half-power beamwidth of this antenna?

(b) What is the front to back ratio of this antenna in dB?

(c) In which direction is the peak power radiation of this antenna?

(d) If a plot of the radiation pattern of a 300 MHz isotropic antenna were added to this same plot, where would the two plots intersect? How do you know this?

26. What is the concept of "reciprocity" in regards to antennas?

27. (*) Elmer is setting up a Wifi router in his home, which is located in the center of his house. He bought some antennas from the store, which include a 2.4 GHz broadside helical antenna (one where the length is much longer than the diameter), a 2.4 GHz log-periodic and a 0.8 GHz dipole antenna. Which is the best option for Elmer to get coverage over his entire house and why?

28. (*) Elmer is interested in setting up a satellite communication system. He currently has a transceiver and a motorized antenna mount (a mount that can change an antenna's orientation). He now needs to decide whether to buy a yagi-uda, dipole, or monopole antenna for his system. Which is the best choice and why?

29. Define "gain" as it relates to antennas. (Be sure to include enough information in your answer to differentiate it from similar antenna concepts).

30. What is the Radar Equation generally used to calculate? Write down the radar equation and define all its variables.

31. What is Impedance? How much impedance does a 5.0Ω resistor have when a 60Hz AC current is applied?